MILITARY SPECIFICATION
FOR
DIAPHRAGM ASSEMBLY

This amendment forms a part of Military Specification MIL-D-21523A (Wep), 23 June 1961, and has been concurred in by the Bureau of Naval Weapons.

Page 5, paragraph 3.3.3: Delete and substitute:

"3.3.3 Rapidly Applied Pressure - With 6 inches of salt water (3 percent salt by weight) maintained at a temperature of +28 ± 2 degrees Fahrenheit on the upper surface of the diaphragm assembly, the diaphragm shall withstand a differential air pressure of 15 ± 1/2 psi applied against the upper surface within 15 to 18 seconds. After the 15 psi limit is reached, the air pressure shall be immediately reduced to 0 psi differential and a differential air pressure of 10 ± 1/2 psi shall be applied against the lower surface within 15 to 18 seconds. After the 10 psi limit is reached, the air pressure shall be immediately reduced to 0 psi differential. The deformation as measured within 1/2 hour after removal of the salt water shall not exceed 1-3/4 inches from the nominally flat surface and thereafter the diaphragm shall be capable of meeting all performance requirements of this specification."

Page 6, add paragraph 3.3.4.1:

"3.3.4.1 Bilateral Flexure at 28 Degrees - With 6 inches of salt water (3 percent salt by weight) maintained at a temperature of +28 ± 2 degrees Fahrenheit above the membrane, the diaphragm assembly shall withstand the bilateral flexure specified in 3.3.4, shall show no water leakage, no permanent deformation of the membrane in excess of 3 inches from the nominally flat surface, and thereafter shall be capable of meeting all performance requirements of this specification."

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:47Z
Check the source to verify that this is the current version before use.

Page 9, paragraph 4.2.3.1: Delete and substitute:

"4.2.3.1 Inspection Lot Sample - Five diaphragm assemblies shall be selected from each succeeding lot of 60, and each shall be tested in accordance with 4.6.2.2 and 4.6.2.3. Failure of any diaphragm during these tests shall be cause for rejection of the lot."

Page 11, paragraph 4.5.1 line 4:

Delete "3.3.1 and 3.3.2" and substitute "3.3.1, 3.3.2, and 3.3.3."

Page 12, add paragraph 4.5.4.1.3:

"4.5.4.1.3 Rapidly Applied Pressure Test - Each diaphragm assembly shall be preconditioned for 3 hours at a temperature of +28 ± 2 degrees Fahrenheit and shall then be subjected to a pressure test to meet the requirements of 3.3.3."

Page 13, add paragraph 4.6.1.6:

"4.6.1.6 Bilateral Flexure Test at 28 Degrees - To meet the requirements of 3.3.4.1, one diaphragm assembly, preconditioned for 3 hours at a temperature of +28 ± 2 degrees Fahrenheit and mounted in a suitable test fixture, shall be covered with 6 inches of salt water maintained at a temperature of +28 ± 2 degrees Fahrenheit. An air pressure of 30 psig shall then be applied simultaneously against the upper and the lower surface of the diaphragm. The diaphragm shall then be subjected to 50 cycles of bilateral flexure induced by alternately varying the air pressure against the upper surface from 25 to 40 psig at a rate to produce not less than 1 cycle of bilateral flexure per minute."

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:47Z
Check the source to verify that this is the current version before use.

Page 13, paragraph 4.6.2.1: Delete.

Page 13, paragraph 4.6.2.2 line 2: Delete "from the combined 3 sublots."

Custodian

Navy - Bureau of
Naval Weapons

Preparing Activity

Navy - Bureau of
Naval Weapons

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:47Z
Check the source to verify that this is the current version before use.